Human Resources Candidate Ranking

Final Technical Report

A two-stage NLP ranking study of graded candidate relevance, duplicate-safe model selection, and recruiter-guided relevance feedback.

1. Executive Summary

This project develops an interpretable candidate-ranking system for two recruiter search intents: “aspiring human resources” and “seeking human resources.” The operational goal is not to predict hiring success. It is to order the available profiles so that the strongest textual matches appear early, where a recruiter is most likely to inspect them.

The source file contains 104 candidate rows, but exact repetition reduces the evidence to 53 unique profile groups. That distinction is central: duplicate rows are not independent candidates or independent human judgements. All model selection therefore uses duplicate-safe grouping, and raw IDs are retained only for traceability and presentation.

Human relevance is graded on a 0-3 scale. Grade 0 means irrelevant to the HR search intent; grade 1 is weak or adjacent; grade 2 is clearly HR-relevant; and grade 3 is a strong/direct match to the aspiring/seeking objective. The labelling exercise showed that HR keyword presence alone is not sufficient: senior HR roles, HR-adjacent roles, and generic seeking/student language can all require different grades depending on intent alignment.

Stage 1 converts title/query text into word TF-IDF, character TF-IDF, BM25, Jaccard, containment, HR-overlap, and intent-overlap evidence. A five-fold GroupKFold protocol evaluates Ridge scoring models with NDCG@10 as the primary metric and MAP/MRR as supporting measures. A broad sensitivity screen covers 192 configurations, including 64 Porter2-stemmed variants. The final retained normalized lexical family selects text-only Ridge with alpha 10 at mean NDCG@10 = 0.9596; adding connection count lowers the corresponding mean to 0.9562.

A matched BM25-versus-TF-IDF comparison has a small observed mean difference (+0.0118) but wide uncertainty: p = 0.46, 95% CI [-0.020, +0.043], with 12 wins, 18 ties, and 10 losses for BM25. The evidence does not establish BM25 superiority. Stage 2 then applies Rocchio relevance feedback and blends 65% immutable Stage 1 with 35% feedback so recruiter preference can change ordering without recursively drifting the baseline.

Bottom line: the system is suitable as recruiter search and prioritization support, not as an automated hiring decision engine. The next improvement should be more unique, consistently labelled relevance data rather than model complexity for its own sake.

2. Introduction and Problem Formulation

2.1 Problem Context and Objective

Recruiter search is an information-retrieval problem: a short query must retrieve and order profiles from a finite candidate corpus. In this dataset, the available evidence is sparse - primarily job title, location, and connection count. There are no full resumes, skill histories, interviews, or employment outcomes, so the modelling scope must remain aligned with what the data can actually support.

The two operational queries are “aspiring human resources” and “seeking human resources.” They express both domain (Human Resources) and intent (aspiring/seeking). A useful ranker must therefore distinguish a direct entry-intent match from a profile that is merely HR-related.

2.2 Ranking Formulation and Scope

Classification answers whether each item belongs to a class; ranking answers which item should appear first, second, third, and so on. Two systems can have identical relevant/not-relevant accuracy while producing very different recruiter experiences if one places the best candidates at the bottom. The project therefore learns a relevance score used only to establish order. The score is not calibrated as a probability.

For each query and candidate title, the system computes lexical evidence, learns a relevance score, and sorts candidates from highest to lowest. Because a strong match to either approved HR query should surface, the final Stage-1 group score uses the stronger normalized score across the two queries.

3. Data and Relevance Labelling

3.1 Dataset and Unit of Analysis

Field	Meaning	Role
id	Source-row identifier	Traceability only
job_title	LinkedIn-style title/headline	Primary relevance signal
location	Candidate location	Context/diagnostics
connection	LinkedIn connection count	Optional feature; not core relevance
graded relevance	Final human grade 0-3	Supervision and evaluation target

The 104 source rows collapse to 53 unique profile groups when normalized title, location, and parsed connection count are identical. The duplicate-size distribution is highly uneven: 39 groups occur once, while several profiles occur four to seven times. A random row split could therefore place the same profile in both training and validation, producing leakage.

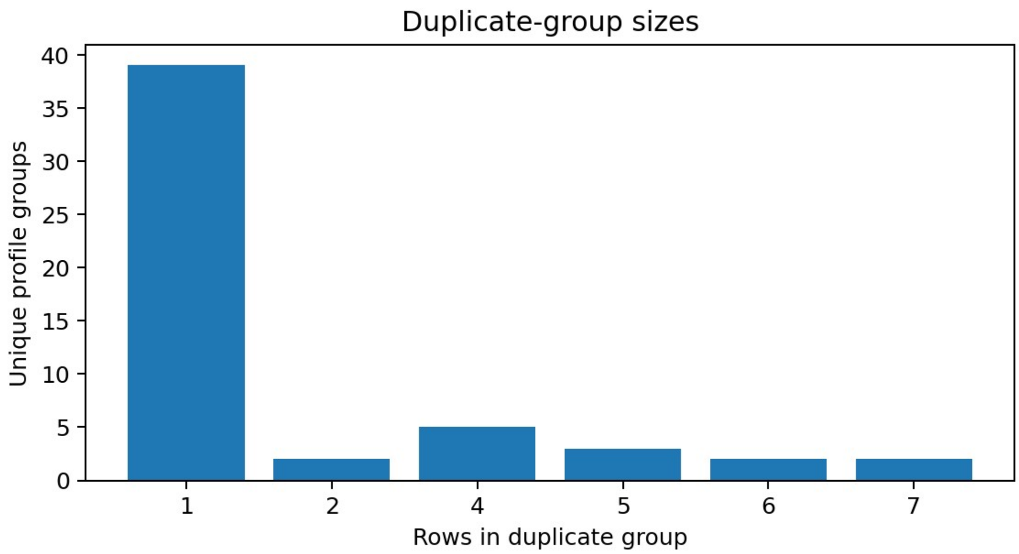

Figure 1. Duplicate-group sizes generated in Notebook 01. The effective modelling sample is 53 unique profile groups.

3.2 Relevance Labelling Methodology

grade	meaning	interpretation
0	Irrelevant	No meaningful HR-search alignment
1	Weak / adjacent	Partial or adjacent evidence; intent mismatch remains
2	Clearly HR-relevant	Clear HR relevance but not the strongest aspiring/seeking match
3	Strong / direct target match	Direct HR relevance with explicit aspiring/seeking/entry intent

Labels were assigned to the unique-profile concept and then propagated to duplicate source rows. This prevents repeated rows from being treated as repeated human judgements. The unique label distribution is 17 grade-0, 2 grade-1, 28 grade-2, and 6 grade-3 profiles. After duplicate propagation, those become 27, 10, 47, and 20 raw rows respectively. In particular, six grade-3 judgements expand to twenty source rows; the evidence remains six independent judgements.

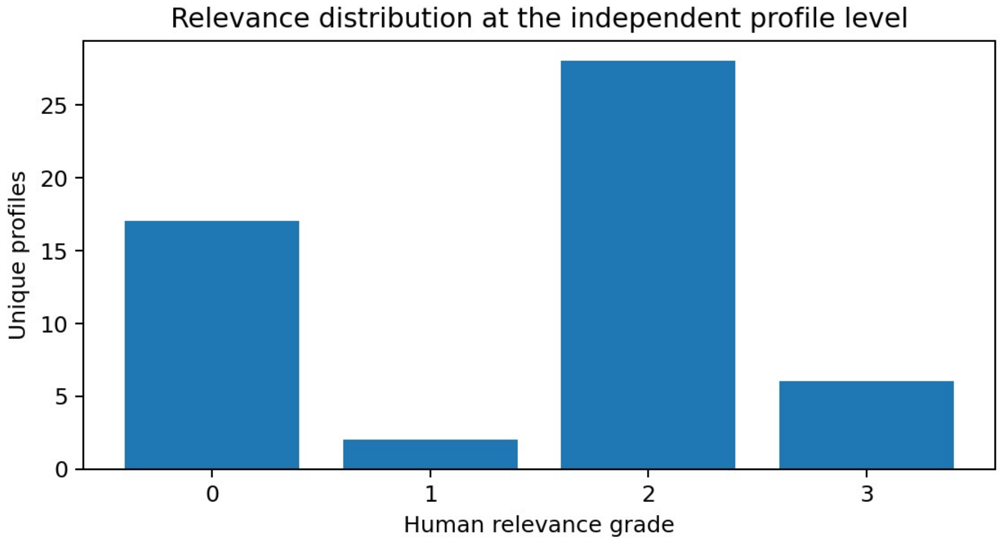

Figure 2. Unique-profile relevance distribution generated in Notebook 01.

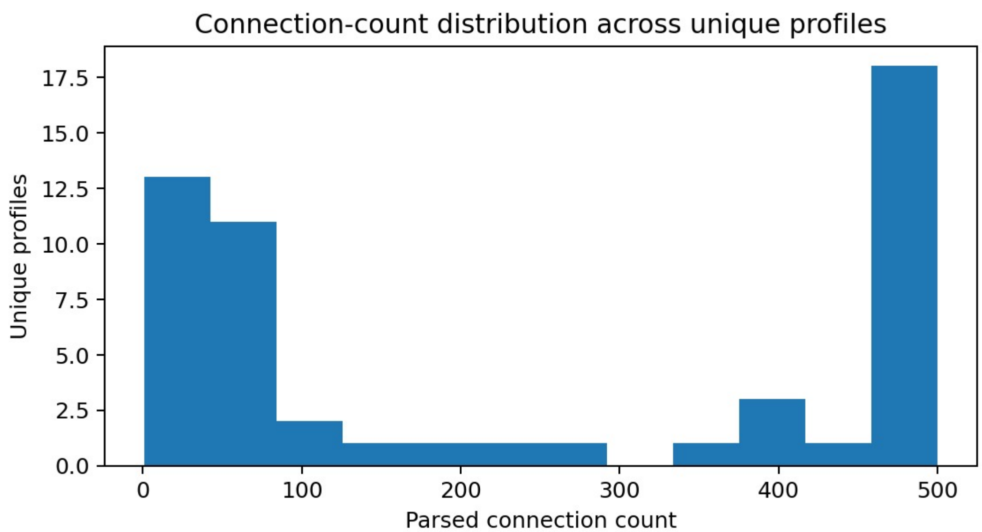

Figure 3. Parsed connection-count distribution across unique profiles, generated in Notebook 01. The skew motivates log transformation when the feature is tested.

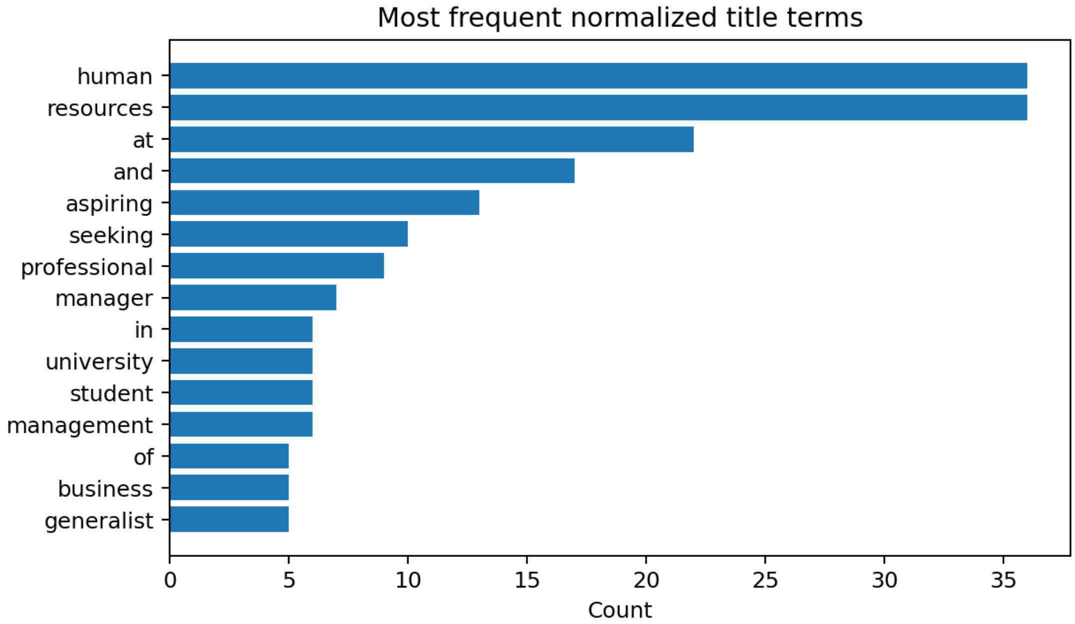

Figure 4. Most frequent normalized title terms, generated in Notebook 01. Human-resources vocabulary dominates, but intent terms are less frequent and therefore useful for distinguishing direct search matches.

Boundary cases required interpretation of business intent rather than mechanical keyword rules. “Aspiring Human Resources Professional” and “Seeking Human Resources Position” are direct grade-3 style matches. Clear HR professionals can be grade 2 when the domain is relevant but the aspiring/seeking intent is less direct. HR-adjacent people-development roles or senior HR leaders can receive lower grades when they do not match the target search objective. Annotation remains subjective and no formal multi-rater reliability estimate is available; this limitation is separate from cross-validation uncertainty.

4. Methodology

4.1 Text Processing and Representation

Normalization lowercases text, standardizes punctuation, expands HR to “human resources,” and normalizes “entry-level” while preserving intent words such as aspiring and seeking. Tokenization turns normalized strings into terms; word n-grams retain short phrases such as “human resources.” A sparse vector stores only non-zero term weights, which is efficient for small text corpora.

TF-IDF combines term frequency (how much a term occurs in a document) with inverse document frequency (how distinctive that term is across the corpus). Cosine similarity compares the direction of the query and candidate vectors. Character TF-IDF uses 3-5 character chunks, making it more tolerant of spelling and morphological variation.

BM25 is a related lexical retrieval score with IDF-like rarity, term-frequency saturation, and document-length normalization. Its k1 parameter controls saturation and b controls length normalization. The pipeline also computes Jaccard similarity, query containment, HR-domain overlap, and intent overlap as transparent evidence.

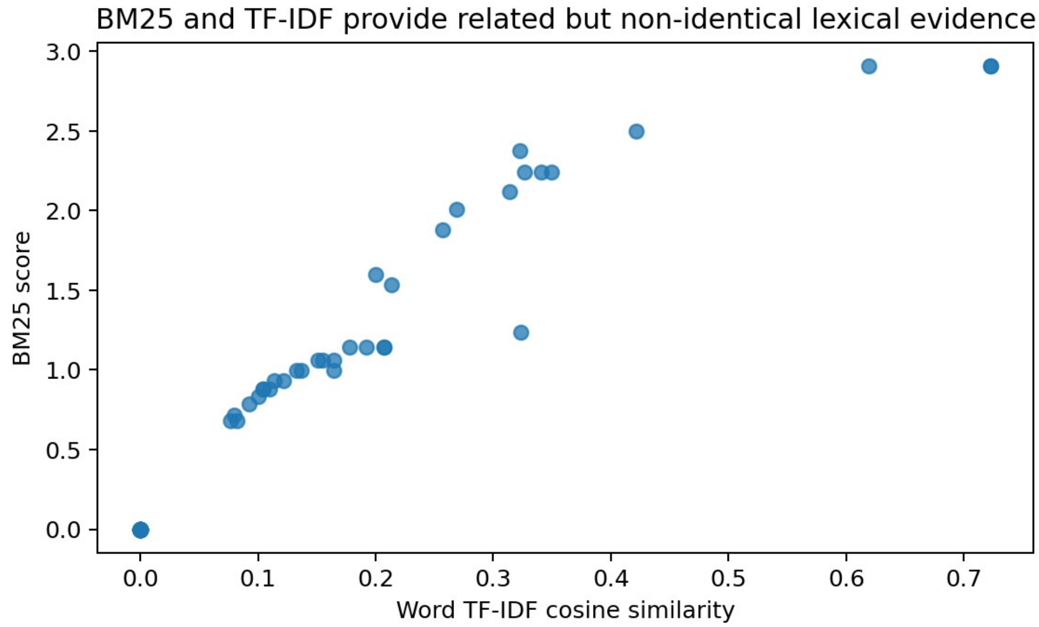

Figure 5. BM25 and word TF-IDF lexical evidence for the aspiring-HR query, generated in Notebook 01.

4.2 Stage-1 Ranking Model

Ridge regression is a linear scoring model with L2 regularization. Standardization puts features on comparable scales, and the alpha hyperparameter controls how strongly large coefficients are penalized. The output is used as a relative relevance score, not a probability. Connection count is log-transformed when tested because raw counts are highly skewed; it is kept optional because network size can act as a seniority, geography, platform-activity, or opportunity proxy rather than true search relevance.

The final retained text-only model uses BM25, word and character TF-IDF, Jaccard, containment, HR overlap, and intent overlap. The model is fit on labelled unique groups for both queries. Query-specific prediction ranges are normalized, then the stronger query score is used as the candidate-level Stage-1 score.

4.3 Stage-2 Relevance Feedback

Rocchio feedback updates the query vector using the original query plus the centroid (average vector) of starred examples and minus the centroid of rejected examples. The project uses alpha = 1, beta = 0.75, and gamma = 0.15. The feedback score is then blended as 0.65 × Stage1 + 0.35 × feedback.

A critical safeguard is that Stage 1 remains immutable. Repeated feedback iterations always blend against the original Stage-1 score rather than recursively blending the previous Stage-2 result. Duplicate IDs are mapped to their shared profile group before feedback centroids are calculated, so repeated source rows cannot multiply one preference signal.

5. Experimental Design and Evaluation

5.1 Model Selection and Cross-Validation

The broad sensitivity screen contains 192 configurations: three preprocessing modes (surface, normalized, Porter2 stemmed), four representation families, four feature bundles, and four Ridge penalties. The Cartesian product gives 3 × 4 × 4 × 4 = 192; because Porter2 is one of the three preprocessing modes, exactly 64 configurations are stemmed. This screen asks whether results are highly sensitive to preprocessing or representation choices.

Dimension	Choices	Count
Preprocessing	surface; normalized; Porter2	3
Representation	word TF-IDF; BM25; word+char TF-IDF;	4

Dimension	Choices	Count
	hybrid	
Feature bundle	retrieval; +overlap; +intent; +connections	4
Ridge alpha	0.1; 1; 10; 100	4

After screening, the final retained family fixes the normalized transparent lexical feature schema and varies five Ridge alphas with/without connection count, producing ten directly comparable candidates. Five-fold GroupKFold is used throughout. With 53 unique groups, each validation fold contains roughly ten or eleven independent profiles, and all duplicates remain in the same fold.

5.2 Ranking Metrics and Statistical Analysis

NDCG@10 is primary because it preserves the full 0-3 relevance grades and gives larger credit when stronger profiles appear earlier. MAP@10 treats grades 2-3 as relevant and averages precision at relevant ranks. MRR looks only at the first relevant result. A zero-relevant evaluation slice is explicitly handled as 0 for AP/MRR rather than producing an undefined division.

ranking	ndcg@5	ndcg@10	map@5	map@10	mrr
score_good	1.0000	1.0000	1.0000	1.0000	1.0000
score_bad	0.6083	0.6083	0.4778	0.4778	0.3333

The BM25-versus-TF-IDF inference uses matched comparisons: each BM25 result is compared with its TF-IDF counterpart under the same evaluation unit. The mean difference estimates the effect size; the confidence interval shows plausible values for that mean difference; the paired t-test evaluates whether a zero mean difference is inconsistent with the observed variation. A p-value is not the probability that one method is better.

6. Results

6.1 Stage-1 Results and Model Selection

Rank	Preprocess	Representation	Feature bundle	Alpha	NDCG@10	MAP@10
1	surface	tfidf_word_char	retrieval	1.0000	0.9641	0.9867
2	normalized	tfidf_word_char	retrieval	0.1000	0.9637	0.9982
3	normalized	tfidf_word_char	retrieval	1.0000	0.9637	0.9982
4	porter2	tfidf_word_char	retrieval	1.0000	0.9636	1.0000
5	normalized	hybrid	retrieval	1.0000	0.9620	0.9909
6	surface	tfidf_word_char	retrieval	0.1000	0.9609	0.9867
7	porter2	tfidf_word_char	retrieval	10.0000	0.9604	1.0000
8	porter2	tfidf_word_char	retrieval	100.0000	0.9604	1.0000

The top broad-screen rows are tightly clustered. Their purpose is to show sensitivity across preprocessing and representation families, not to turn a difference of a few thousandths on 53 unique profiles into a durable winner claim.

Model / checkpoint	NDCG@10	MAP@10	MRR
Normalized word TF-IDF screen	0.9550	1.0000	1.0000
Normalized BM25 screen	0.9527	0.9909	1.0000
Normalized word+character TF-	0.9637	0.9982	1.0000

Model / checkpoint	NDCG@10	MAP@10	MRR
IDF screen			
Normalized BM25+word+character screen	0.9620	0.9909	1.0000
Retained Ridge text-only, alpha=10	0.9596	1.0000	1.0000
Retained Ridge + connections, alpha=10	0.9562	1.0000	1.0000

The broad screen confirms that several lexical formulations perform similarly at the top of the list. The highest raw broad-screen mean is about 0.9641 NDCG@10 for a surface word+character TF-IDF configuration. The normalized word+character and normalized hybrid screens are close at 0.9637 and 0.9620. Because the dataset is only 53 unique profiles, those tiny gaps should not be over-interpreted as stable model-family winners.

Inside the retained normalized feature family, text-only Ridge alpha 10 is best at 0.9596 mean NDCG@10. Adding connection count lowers mean NDCG@10 to 0.9562. Weak regularization performs worse, while MAP/MRR saturate at 1.0 across many strong variants, demonstrating why graded NDCG is the more discriminating primary metric.

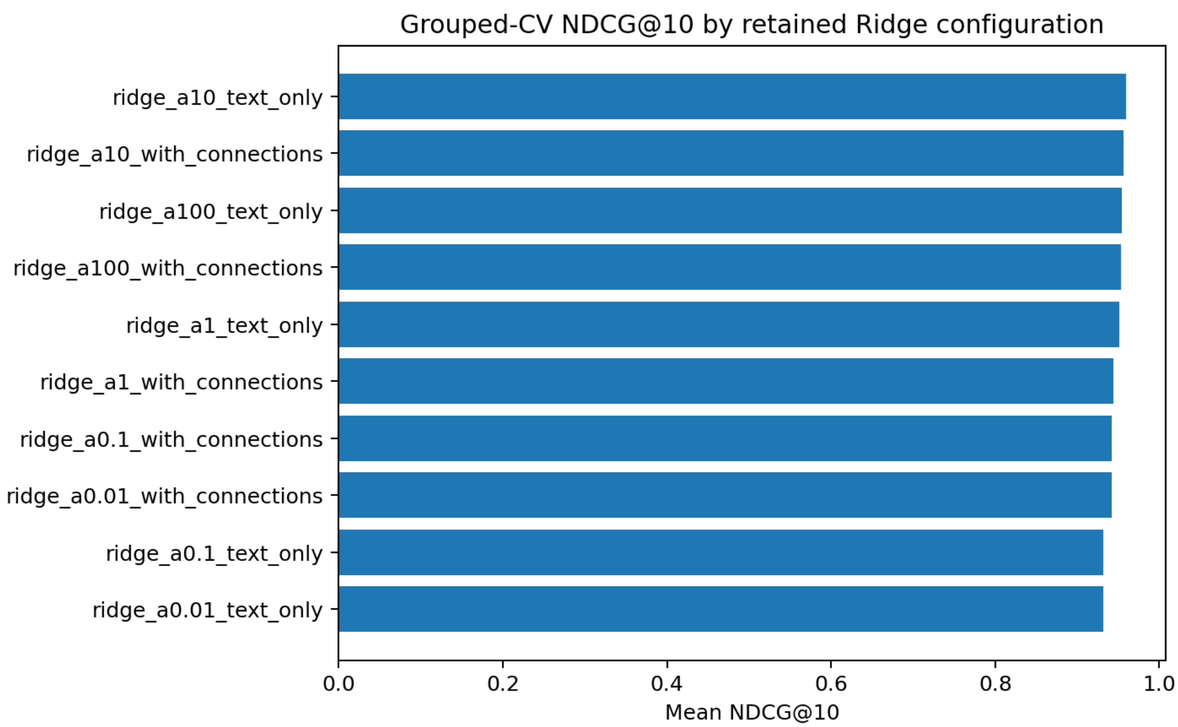

Figure 6. Mean NDCG@10 for the ten retained Stage-1 configurations, generated in Notebook 02.

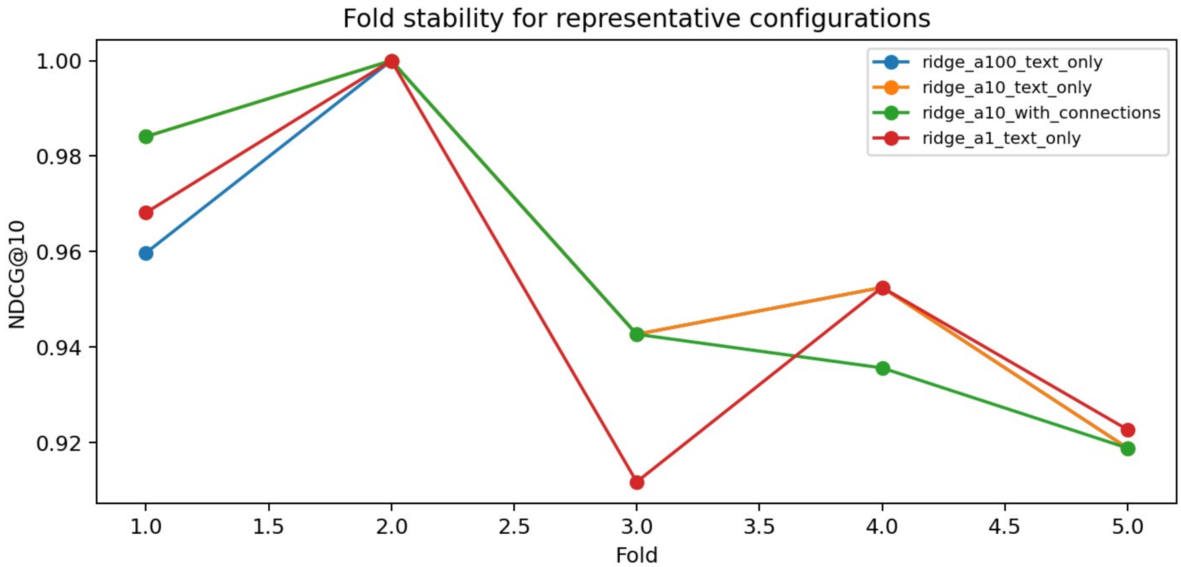

Figure 7. Fold-level NDCG@10 stability for representative retained configurations, generated in Notebook 02.

quantity	value
Mean paired difference (BM25 - TF-IDF)	0.0118
Paired t-test p-value	0.4600
95% CI lower	-0.0200
95% CI upper	0.0430
Wins	12.0000
Ties	18.0000
Losses	10.0000

The observed BM25 mean is slightly higher in the matched comparison, but p = 0.46 and the 95% interval crosses zero. The 12/18/10 win/tie/loss split is also mixed. The appropriate conclusion is therefore similarity with uncertainty, not BM25 superiority.

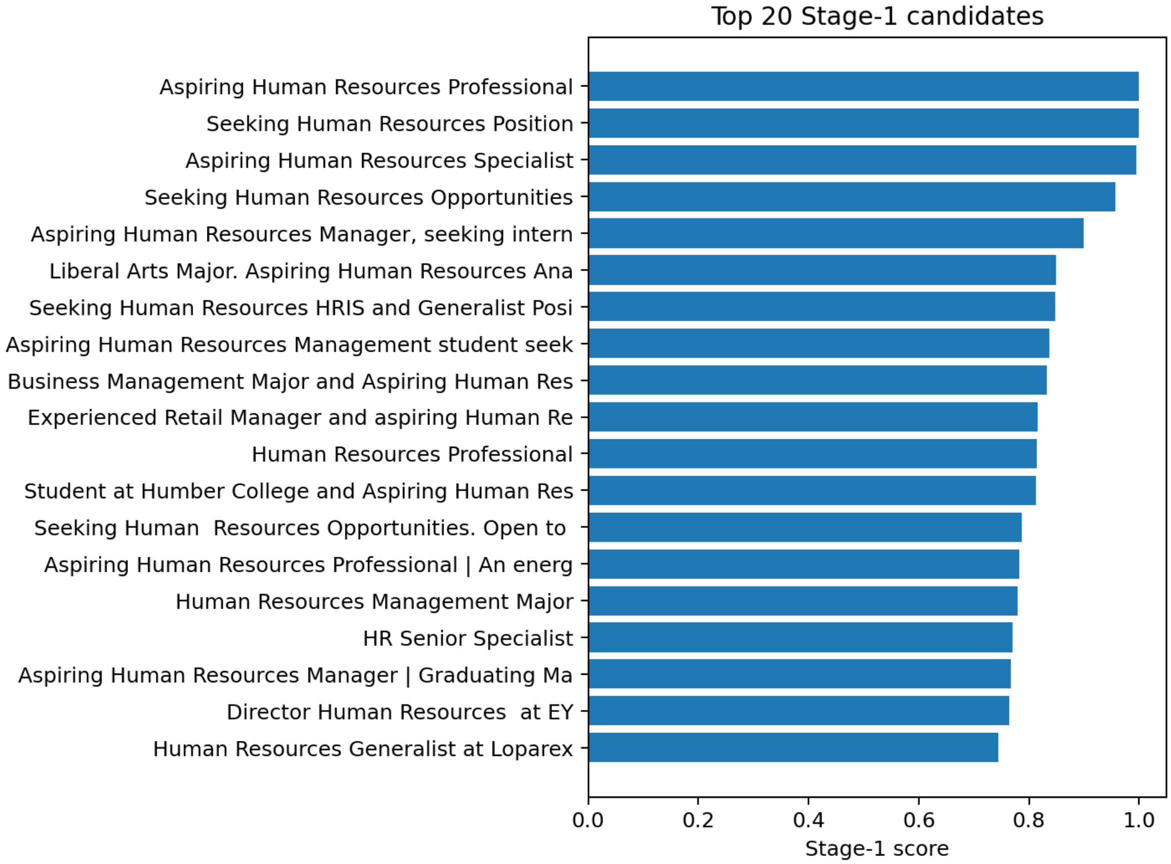

Figure 8. Final Stage-1 top 20 unique profiles after fitting the selected retained model on all labelled groups, generated in Notebook 03.

6.2 Stage-2 Reranking Results

Rank	ID	Job title	Grade	Score
1	99	Seeking Human Resources Position	2.0000	1.0000
2	97	Aspiring Human Resources Professional	2.0000	1.0000
3	3	Aspiring Human Resources Professional	3.0000	1.0000
4	6	Aspiring Human Resources Specialist	3.0000	0.9954
5	28	Seeking Human Resources Opportunities	3.0000	0.9568
6	73	Aspiring Human Resources Manager, seeking internship in Human Resources.	3.0000	0.8989
7	79	Liberal Arts Major. Aspiring Human Resources Analyst.	2.0000	0.8490
8	10	Seeking Human Resources HRIS and Generalist Positions	3.0000	0.8474
9	27	Aspiring Human Resources Management	3.0000	0.8374

Rank	ID	Job title	Grade	Score
		student seeking an internship		
10	72	Business Management Major and Aspiring Human Resources Manager	2.0000	0.8318

The top-ranked titles provide a qualitative sanity check: direct aspiring/seeking HR titles and clearly HR-relevant profiles dominate the leading positions, while unrelated careers fall much lower. This does not replace quantitative validation, but it helps detect obviously nonsensical scoring behavior.

With candidate ID 3 starred as a positive example, Rocchio feedback changes the language emphasis while preserving 65% of the Stage-1 baseline. The reranking moves some “aspiring Human Resources Professional” profiles upward and can move generic or differently phrased HR profiles downward. This is the intended behavior: feedback personalizes the current search rather than globally retraining the Stage-1 model from one click.

Figure 9. Stage-2 top 20 after Rocchio feedback, generated in Notebook 03.

ID	Job title	Stage-1 rank	Stage-2 rank	Change
76	Aspiring Human Resources Professional | Passionate about helping to create an inclusive and engaging work environment	28	19	9
1	2019 C.T. Bauer College of Business Graduate (Magna Cum Laude) and aspiring Human Resources professional	27	20	7

ID	Job title	Stage-1 rank	Stage-2 rank	Change
74	Human Resources Professional	12	6	6
70	Retired Army National Guard Recruiter, office manager, seeking a position in Human Resources.	21	26	-5
101	Human Resources Generalist at Loparex	20	24	-4
84	Human Resources professional for the world leader in GIS software	33	29	4
78	Human Resources Generalist at Schwan's	23	27	-4
66	Experienced Retail Manager and aspiring Human Resources Professional	11	8	3

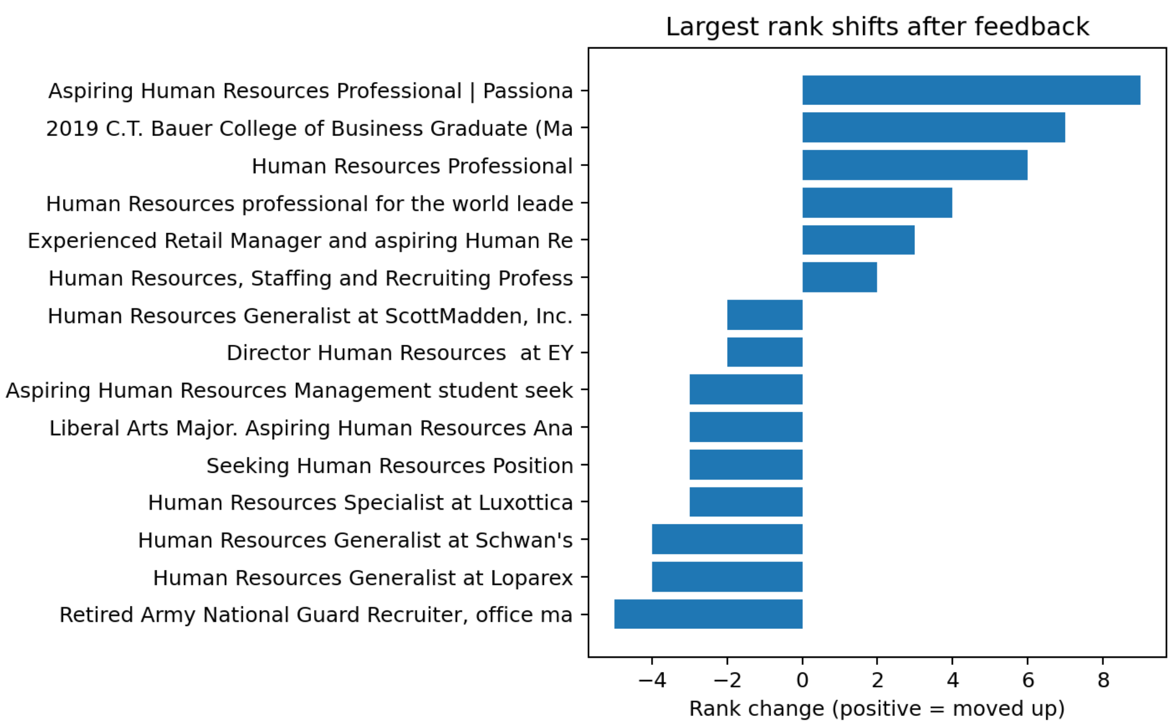

Figure 10. Largest Stage-2 rank movements after feedback, generated in Notebook 03. Positive values indicate movement upward.

7. Discussion

7.1 Interpretation of Findings

The first important result is methodological: ranking is the correct formulation. The business value lies in bringing stronger candidates forward, not merely assigning correct binary labels. Graded relevance improves that objective by distinguishing direct target matches from profiles that are useful but less aligned.

The second result is that transparent lexical methods are already strong for this dataset. Broad screening does not reveal a decisive need for semantic transformers, and the BM25-versus-TF-IDF uncertainty reinforces the same lesson. The next model need not be more complex simply because more complex architectures exist.

The third result is that the labelled evidence, not the algorithm, is the main constraint. With 53 unique groups and subjective grades, tiny cross-validation differences are fragile. More independent annotations and prospective recruiter outcomes would provide more information than further micro-tuning on the same sample.

7.2 Robustness and Methodological Safeguards

- Duplicate groups never cross training/validation folds.
- Zero-relevant metric slices return defined values instead of NaN.
- Connection count is optional and excluded from the selected compact model.
- Stage-2 feedback maps raw IDs to unique profile groups.
- Stage 2 always blends against the immutable Stage-1 baseline.
- Scores are described as relevance/order signals, not hiring probabilities.

8. Limitations and Future Work

8.1 Limitations

- Only 53 unique labelled profile groups support model development and evaluation.
- Repeated raw rows do not increase the independent sample size.
- Relevance grades are subjective; no formal inter-rater agreement statistic is available.
- The same small dataset is reused for model development, so repeated experimentation can overfit the evaluation process.
- Most predictive evidence is job-title text; richer resume, skill, and experience data are absent.
- Lexical methods can miss semantic equivalence that does not share surface vocabulary.
- Location and connection-related data can function as socioeconomic or opportunity proxies; fairness is not established.
- Offline NDCG/MAP/MRR do not prove recruiter productivity, shortlist quality, or hiring outcomes.

8.2 Future Work

- Collect more unique profiles with the same graded rubric and preserve query context.
- Use a second reviewer on a subset and measure agreement; record short rationales for difficult labels.
- Create harder negative examples and a prospective holdout set that is not reused during tuning.
- Pilot with recruiters and measure time to first useful candidate, top-k acceptance, profiles reviewed per shortlist, no-match/abandonment, and satisfaction.
- Consider query-specific labels as search intents diversify.
- Evaluate semantic embeddings only after more supervision is available and compare them against this lexical baseline under the same duplicate-safe protocol.
- Monitor fairness and proxy effects before using any ranking signal in consequential decisions.

9. Conclusion

The project demonstrates a complete, defensible ranking workflow on a small HR candidate corpus: define graded relevance, collapse duplicates to the correct unit of analysis, convert title/query text into interpretable lexical evidence, tune under grouped cross-validation, evaluate with ranking-specific metrics, and add controlled recruiter feedback as a second stage. The selected compact Stage-1 family performs strongly without connection count, while the broader comparisons show that tiny lexical-method differences should not be promoted into unsupported winner claims.

The system should therefore be deployed, if at all, as a human-supervised search aid. Its strongest near-term improvement is a better evidence loop - more unique labels, clearer annotation consistency, prospective evaluation, and recruiter usage data - rather than additional model complexity for its own sake.